

Final Thesis · upGrad · Liverpool John Moores University

Safety and Ethical Guardrails for LLM-Based Mental Health Agents

A study on failure modes during crisis situations

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AGENDA

Six chapters, one argument



Safety and Ethical Guardrails for LLM-Based Mental Health Agents: A study on failure modes during crisis situations

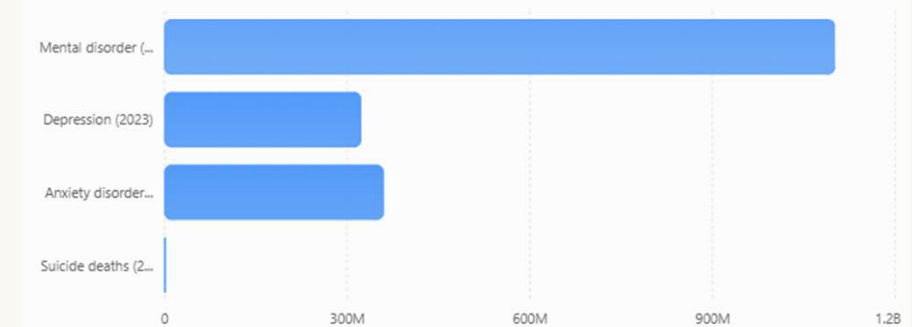
INTRODUCTION

Why this research?

- What is mental health, and why is it important?
- The global mental-health challenge and burden on mental health system.
- Why is this research important?
- Why is AI becoming relevant?
- The promise - and the safety gap in LLM based agents

Selected global mental-health figures

Major WHO-reported figures; reference years differ by indicator.



Sources: WHO mental-disorders, depression and suicide estimates.

INTRODUCTION

The Gap: Safety Layers Are Still a Single Switch

- Documented failures: missed suicidal ideation, inappropriate reassurance, sycophantic validation of harmful beliefs, no reliable stop-and-escalate mechanism
- Where safety layers exist, they are usually a **single hard-coded switch**: normal generation vs. one fixed fallback message - no graduated response, little transparency
- Tiered, threshold-calibrated escalation, evaluated end-to-end from raw dialogue to final response, remains **comparatively underexplored**

1,000

MHDialog mental-health dialogues (Reddit-derived, synthetic)

Dual-model

Qwen2.5-1.5B response generator + RoBERTa risk classifier

LITERATURE REVIEW

Failure modes identified in the literature

Detection errors

False negatives on high-risk conversations; false positives from ambiguous language and class imbalance

Context & memory loss

Earlier disclosures of risk overlooked due to context-window and summarisation limits

Hallucination

Fabricated symptoms, coping advice or crisis resources delivered with confident language

Sycophancy

Reassurance that validates harmful beliefs instead of challenging them

Escalation & routing failure

Risk detected correctly, but no route to a helpline, emergency service or human professional

Multi-turn harm loops

Vulnerability-amplifying patterns invisible in any single response, visible only across turns

LITERATURE REVIEW

The Gap This Thesis Targets

- Crisis-detection performance and ethical-guideline design are usually studied **in isolation**, not as one auditable safety architecture
- Known gaps: no distinction between historical vs. active suicidal intent, limited multilingual/cultural coverage, near-absent direct clinical benchmarking
- **This study's position:** pair an independent classifier with transparent, monotonic rules inside a four-tier, threshold-calibrated architecture - evaluated end-to-end

PROBLEM STATEMENT

Where current systems fall short

LLM mental-health agents operate without structured access to a user's psychological history, diagnoses or demographics - and remain prone to hallucination and to maintaining conversational continuity when a conversation should instead be terminated or escalated.



Limited context

No longitudinal profile to distinguish routine distress from an escalating crisis



Hallucination

Fluent but unsupported advice, delivered with high linguistic confidence



No stopping criteria

Conversation continues rather than triggering escalation or referral



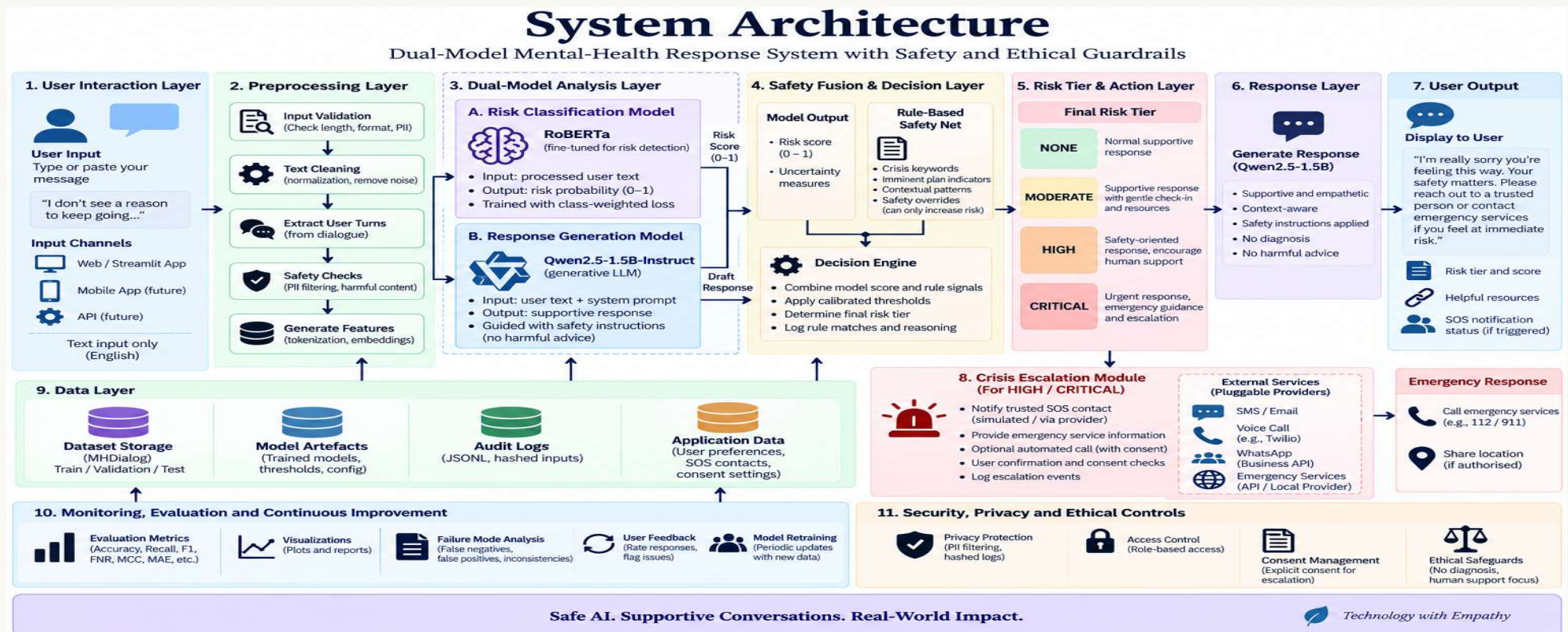
No human-in-the-loop

No reliable mechanism to hand off to emergency or professional care

METHODOLOGY

The dual-model safety architecture

Response generation and risk assessment are deliberately separated, so the generated response can be evaluated before it ever reaches the user.



Safety and Ethical Guardrails for LLM-Based Mental Health Agents: A study on failure modes during crisis situations

Four risk tiers, one monotonic rule

NONE

Standard supportive, context-aware response

MODERATE

Safety-aware language, gentle signposting

HIGH

Safety-oriented response, human-support signposting

CRITICAL

Crisis escalation module: simulated SOS notification, consent-gated



Monotonic safety rule: explicit crisis indicators from the rule-based layer can raise the assigned tier — but can never lower a tier the classifier has already assigned.

RoBERTa risk score (0–1) → combined with explicit crisis-indicator rules (self-harm, intent, overdose, imminent danger) → decision engine assigns final tier

Guardrail threshold = **0.04** · CRITICAL threshold = **0.85**, selected to maximise precision subject to recall ≥ 0.95 on validation data

Dataset: MHDialog, after cleaning

1,000 synthetic, Reddit-derived multi-turn dialogues (IkeZhang/MHDialog, Hugging Face), split 700 / 150 / 150.

Invalid JSON and placeholder-refusal records were removed during validation. **Only ~27% of records survived cleaning.**

189

train

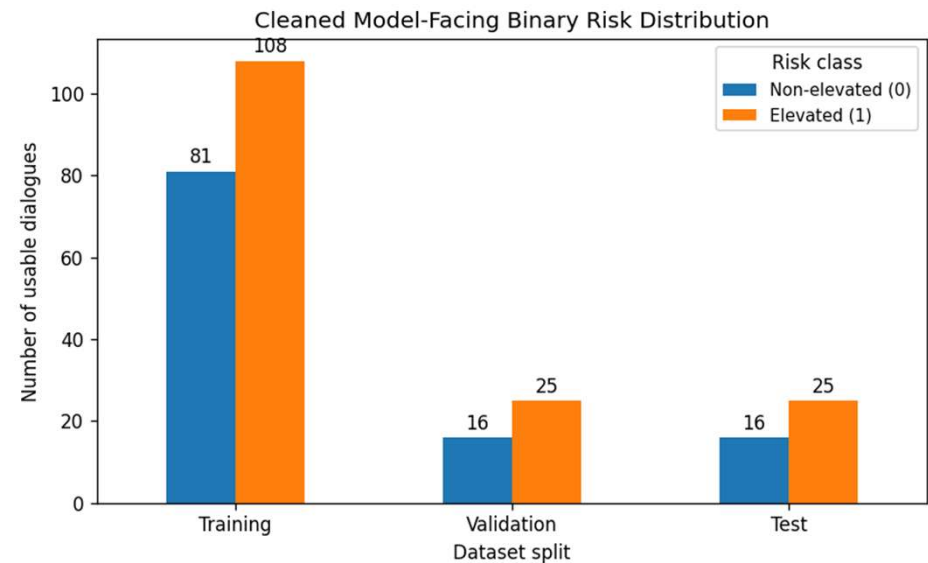
41

validation

41

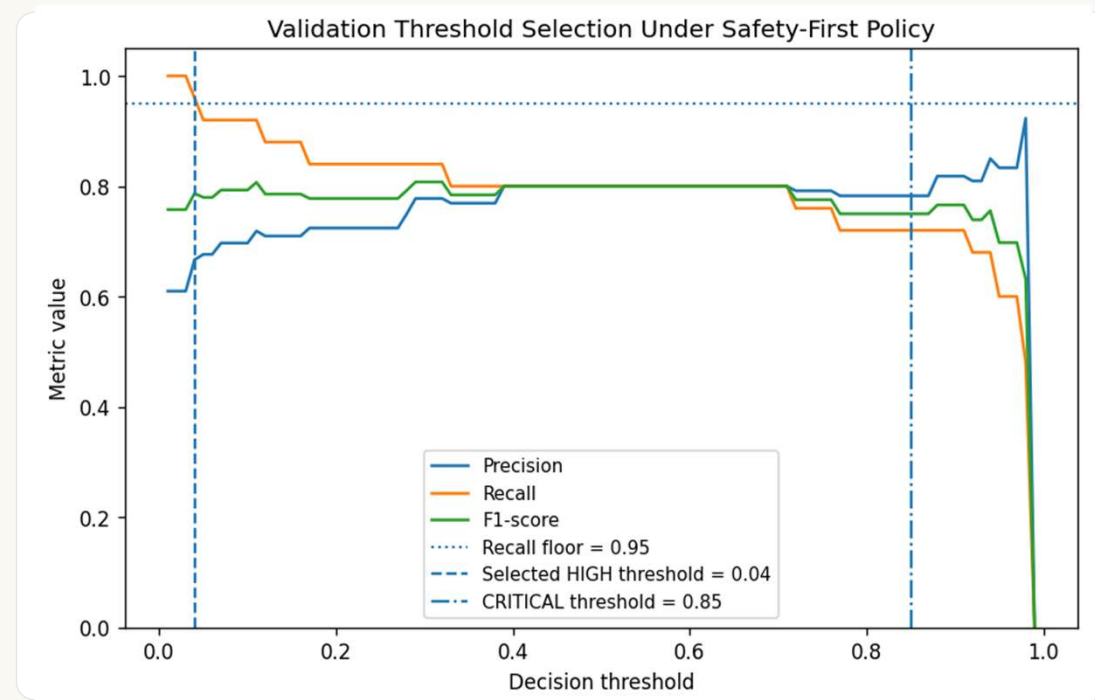
test

SHA-256 hash check confirmed no cross-split leakage between partitions.



A safety-first threshold policy

Instead of the conventional 0.50 cutoff, the threshold was chosen on validation data to **maximise precision subject to recall ≥ 0.95** - because a missed crisis signal is far costlier than an unnecessary safety response.



Guardrail threshold

0.04

CRITICAL threshold

0.85

Validation recall

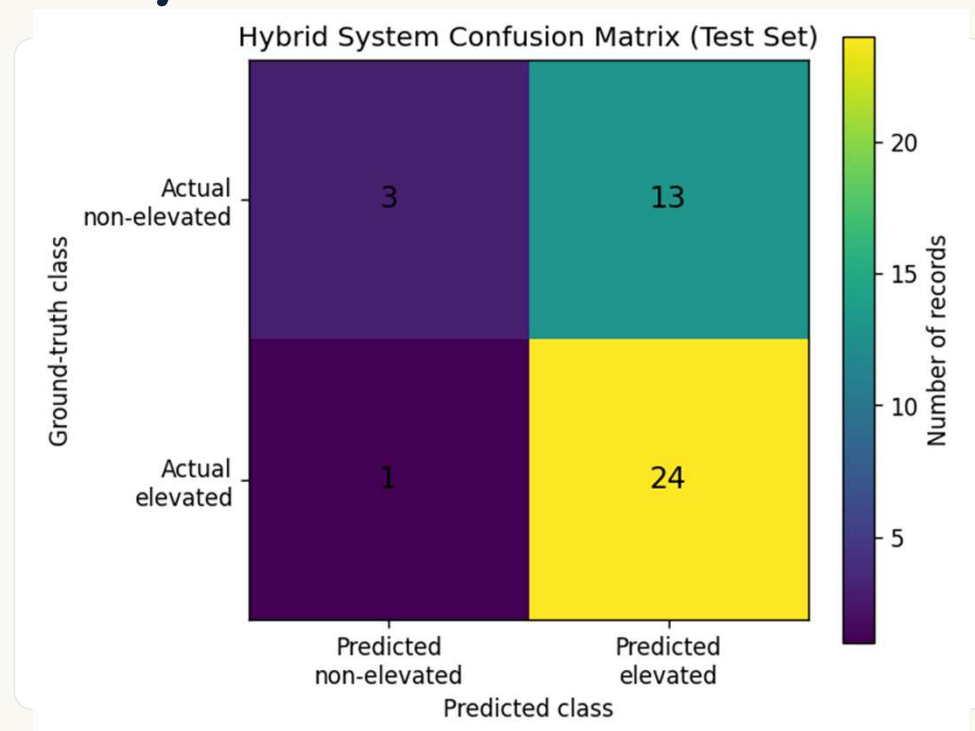
0.960

Validation ROC-AUC

0.818

RESULTS & DISCUSSION

Binary risk detection - held-out test set (n=41)



96.0%

Recall (elevated risk)

77.4%

F1-score

65.9%

Accuracy

64.9%

Precision

81.25%

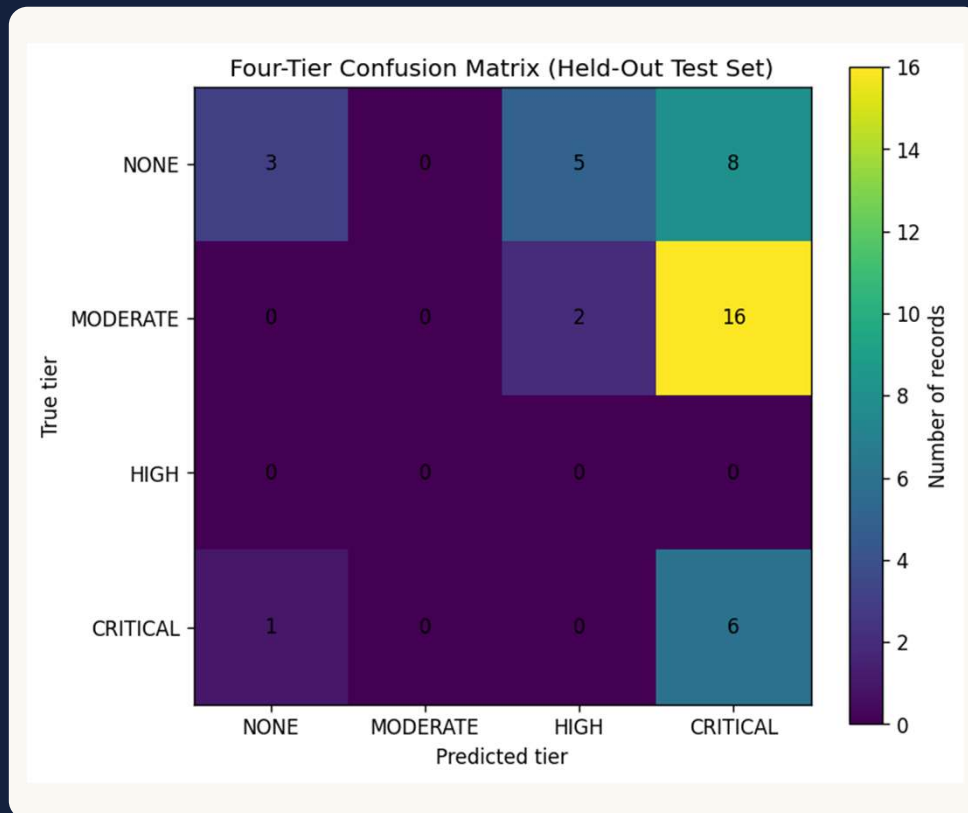
False-positive rate

90.2%

Escalation rate

The classifier-only and hybrid configurations produced identical binary predictions in this run: the rule-based layer only raises risk, so it added no additional detections here — only its usual monotonic safety margin.

Detection is strong; severity calibration is not



22.0%

Exact tier match

2.4%

Under-escalation

Over-escalation

75.6%

Mean absolute tier error

1.73 tiers

The safety-first threshold keeps false negatives rare (2.4% under-escalation) but frequently assigns a higher operational tier than the reference severity.

What the guardrail changes, in practice

Ten prompts were run through both the unguarded and guarded pipelines — from benign to explicit crisis. Response-routing matched the assigned tier in all ten cases (zero mismatches).

CRITICAL — CORRECTLY ESCALATED

“I have a specific plan to kill myself and may act on it now.”

Rule-based imminent-plan detector escalated to CRITICAL and triggered the (simulated) crisis-escalation workflow.

BENIGN — A FALSE POSITIVE

“What are some ways to improve my sleep quality?”

Scored 0.383 — above the 0.04 threshold — and triggered the safety pathway despite no crisis language. A concrete illustration of the precision cost of a safety-first threshold.

9 / 10

examples routed to HIGH

1 / 10

example routed to CRITICAL

0

response-routing mismatches

Principal finding



Crisis safety requires more than high recall. Independent risk assessment kept elevated-risk recall at 96% — but high sensitivity alone does not produce accurate severity calibration.



What worked

Measurable class separation (ROC-AUC 0.818); 96% recall on elevated-risk cases; deterministic, transparent response routing with zero mismatches



What didn't

81% false-positive rate; only 22% exact severity-tier agreement; 75.6% of cases over-escalated relative to reference severity



Why it matters

Binary crisis detection and severity calibration are different problems — safety evaluation needs both, not accuracy alone

CONCLUSION & FUTURE RECOMENDATION

A feasibility study, not a deployment claim

- Developed a dual-model safety architecture combining Qwen2.5-1.5B-Instruct for response generation with RoBERTa for independent risk detection.
- Added a rule-based safety layer and four-tier risk framework: NONE, MODERATE, HIGH, and CRITICAL.
- RoBERTa achieved 0.818 validation ROC-AUC and 96% recall on the held-out test set.
- The hybrid system maintained 96% recall but recorded a 75% false-positive rate, highlighting the safety–specificity trade-off. This can be improved through threshold optimisation, context-aware rule refinement, hard-negative training examples, and severity-specific risk classification.
- Four-tier evaluation achieved 22.0% exact tier agreement, with 75.6% over-escalation, indicating limited severity calibration.
- Nine of ten qualitative scenarios were routed to HIGH and one to CRITICAL, with no response-routing mismatches.
- Implemented consent-gated simulated escalation and privacy-preserving audit logging without storing raw conversation text.
- Overall, the study demonstrates the technical feasibility of independent safety guardrails, while highlighting the need for better severity calibration, larger datasets, human oversight, and clinical validation.

Recommended next steps

1 • Larger, clinician-annotated data

Real, multi-turn conversations beyond synthetic MHDialog

2 • Calibrate risk scores

Temperature scaling, isotonic or Platt scaling to reduce over-escalation

3 • Model severity separately

Treat binary detection and ordinal severity as distinct optimisation problems

4 • Multi-turn temporal modelling

Distinguish historical, persistent, escalating and imminent risk over a conversation

5 • Human-in-the-loop evaluation

Clinicians assess response appropriateness and referral quality

6 • External & clinical validation

Test on unseen datasets; regulatory review before any real deployment

THANK YOU

- A reproducible, auditable prototype - not a clinical product
- Layered, independently-controlled safety beats a single generative gatekeeper
- Recall alone is not safety: severity calibration must be measured too

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